
Beamforming for Imaging: A Brief Overview

Jan Egil Kirkebø



Introduction

In *echo-imaging* a wavefield is emitted into a medium, and through scattering and reflection an image can be formed from the received signals. Even though echo-imaging is a relatively new concept in the technological world, it is nothing new under the sun in the natural world. Many animals, such as dolphins, bats and oilbirds, use echo-location (also referred to as biosonar) to locate and identify objects. A high-frequency ultrasonic pulse is emitted, which is reflected off of surrounding objects, and the time-delay before the echos return can be used to determine the distances to the surrounding objects. Figure 1 illustrates how a bat performs echo imaging as it approaches its target. Bats typically emit a *chirp*, i.e. a waveform where the frequency varies as a function of time. The distance to the target can then be determined from the two-way travel time of this emitted pulse. The bat's "vision" depends on the frequency range and the temporal variation of the frequency of the chirp. During the various stages of the capture process bats typically emit five different types of chirps [1], varying in frequency range and time duration.

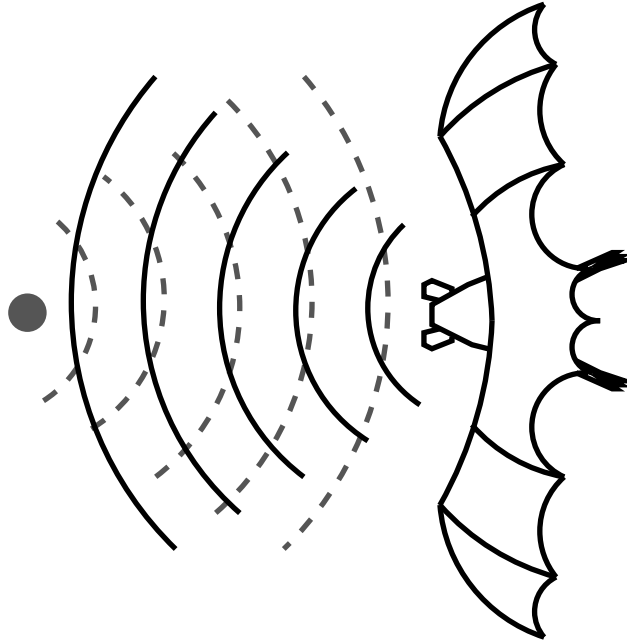


Figure 1: Illustration of a bat performing echo-imaging. A high-frequency ultrasound pulse is emitted, which is reflected off of surrounding objects.

The principles of imaging systems are basically the same as for biosonar. The process of imaging in sonar or medical ultrasound can roughly be described by the following four steps:

1. The aperture is excited by an electric signal, so that it emits a wavefield into the medium of interest.
2. The emitted wavefield propagates through the medium, experiencing attenuation and diffraction, as well as scattering and reflection.
3. The reflected and scattered wavefield propagates back towards the array of elements, where upon reception the measured wavefield is converted into electric signals.
4. The electric signals transformed from the received echos at the aperture are processed so that they can be used to form an image.

Note that in the case of sonar systems some operate in passive mode, where it is the received signal of the target's self-noise which is analyzed. In these systems the process of imaging can be described by only the last two of the abovementioned steps. In active mode systems, where the imaging process consists of all the abovementioned four steps, the transmit aperture does not necessarily have to coincide with the receive aperture.

Whether it be the air, the body or the ocean, all real media have at least some inherent noise. In imaging everything besides propagating wavefields from the look direction is considered noise. All imaging systems based on the abovementioned four steps suffer in noisy environments, which in most cases can cause significant degradation to the system's performance. To mitigate the effects of noise, and thus improve the performance of the imaging system, it is desirable to increase the received signal-to-noise ratio. One way is through a focusing of the wavefield, which can be employed if the target is near enough (or steered otherwise). There are several ways of achieving focusing, both at transmission and reception:

- The aperture can be curved in space.
- A lens can be placed in front of the aperture.
- The output from each of the elements in an array can be phased, or a time-delay can be imposed. The latter is part of what is referred to as beamforming.

A limiting factor of imaging systems is that it is impossible to focus a wavefield perfectly (at either transmission or reception) using a finite size aperture. This is equivalent to the *time-frequency uncertainty principle* for time

sequences, which gives a lower bound on the time-bandwidth product. In most of the current state of the art imaging systems which require focusing, arrays are preferred since a two-dimensional (2D) or three-dimensional (3D) volume can be scanned by adding a phase or time-delay to the array elements, without the system having to contain moving mechanical parts. Arrays will also be the focus here.

Beamforming can be described as the technique of using an array of transducer elements to focus or steer a wavefield, and can be employed both at transmission and reception. At transmission both the amplitude and the time of excitation are controlled at each element so that propagating waves add up constructively in the focal point, and have as much destructive interference as possible at all other locations. At reception the received signals are weighted and added coherently (phased¹) so that the wavefield from the desired direction is reinforced while it is suppressed as much as possible from all other directions. For conventional beamforming the question of phase control or time-delay reduces to simple geometry, translating the path length to distance travel time. The key principles of beamforming are illustrated in Figure 2. It is, by large, the properties of the emitted and the scattered and reflected wavefield that decide the image quality, and therefore a great deal of attention must be made to beam optimization.

The shape of the beam is usually quantified through the beampattern, which is the angular response of an array to a plane wave. Define the wavenumber vector $\mathbf{k} = 2\pi \cdot \mathbf{s}/\lambda$, where $\mathbf{s}$ is the unit direction vector pointing in the same direction as the plane wave and λ is the wavelength. An array's response to a plane wave in the far-field with wavenumber vector $\mathbf{k}$, assuming omni-directional elements, is simply the spatial Fourier transform over the array's element weights:

$$y(\mathbf{k}) = \sum_{m=0}^{M-1} w_m e^{j\mathbf{k} \cdot \mathbf{x}_m}, \quad (1)$$

where $\mathbf{x}_m$ is the location of the m th array element. This is analogous to the *frequency response* of a finite impulse response (FIR) filter. If all the weights are equal, which is referred to as uniform weighting, the array acts as a spatial (low-pass) moving average filter. Unless otherwise specified, uniform weighting has been employed in all of the examples below.

¹Ideally one would wish to time-delay the various received signals, though this was quite difficult in traditional analog circuitry. However, phased-array beamforming gives a good approximation to time-delay beamforming as long as the narrowband wavefield travel-time across the array is much smaller than the pulse length.

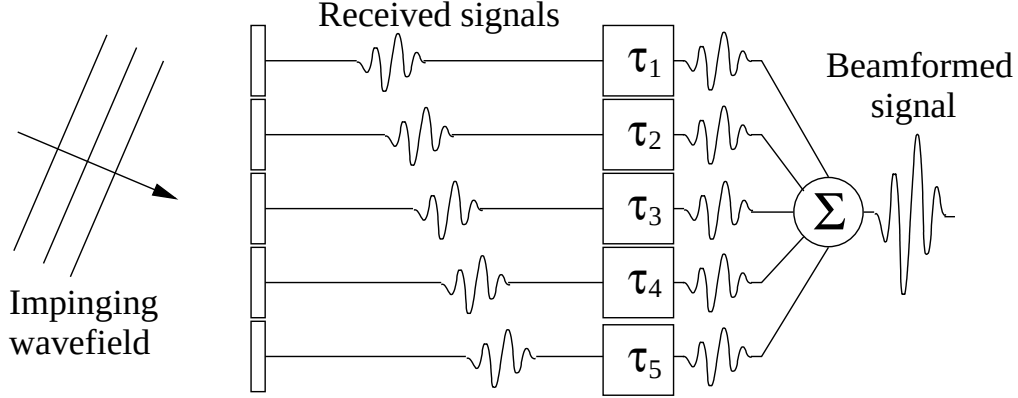


Figure 2: Illustration of the principles of beamforming. A wavefield impinges on an array of elements at some angle. Each element records a time-delayed version of the wavefield. Each of these recordings are then time-delayed, so that they interfere constructively when being summed.

A commonly used array is the one containing M equispaced elements on a line, which is referred to as a *uniform linear array*, having the element placements

$$x_m = \left(m - \frac{M+1}{2}\right) d, \quad m = 1, \dots, M,$$

where d is the interelement distance. The time-delay required for the m th element to steer the beam in the direction θ_0 in the far-field can easily be shown to be

$$\tau_m = \frac{md}{c} \sin \theta_0, \quad m = 1, \dots, M,$$

where c is the speed of sound. The array's response, as defined in (1), then reduces to

$$y(k_x) = \sum_{m=1}^M w_m e^{jk_x \cdot (m-(M+1)/2)d},$$

where k_x is the wavenumber vector in one dimension. For the uniform linear array the array's response is periodic with period $n \cdot 2\pi/(k_x d)$, for some integer n .

In an analogous manner to how *Shannon's sampling theorem* is given as consequence of the periodicity of the frequency response of a linear and time-invariant filter, so does the periodicity of a uniform linear array give rise to an equivalent spatial sampling theorem. To avoid the ambiguities caused by

*grating lobes*² when imaging targets over a 180° sector, the element spacing must be less than half a wavelength, i.e. $d < \lambda/2$.

The beampattern of a 10 element uniform linear array with an interelement distance of $\lambda/2$ is shown on the left hand side of in Figure 3, with the mainlobe and sidelobes pointed out. On the right hand side of Figure 3, the beampattern is shown for an equivalent array having an interelement distance of λ , with the mainlobe and a grating lobe pointed out.

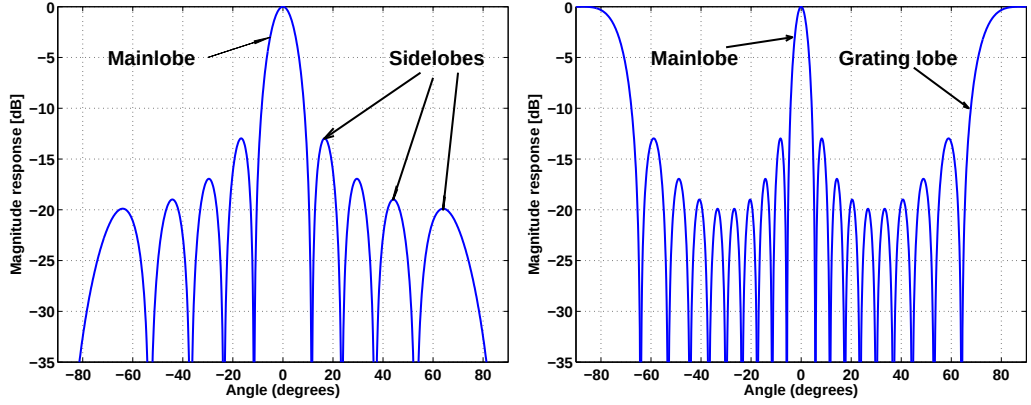


Figure 3: Beampattern of a 10 element uniform linear array with an interelement distance equaling $\lambda/2$ (left) with the mainlobe and sidelobes pointed out, and with an interelement distance of λ (right) with the mainlobe and a grating lobe pointed out.

Three measures of performance related to the beampattern are of particular interest with respect to imaging resolution:

1. The width of the mainlobe relates to the azimuth/lateral resolution of the system, and is largely determined by the size of the aperture. A larger aperture results in a narrower mainlobe, which means an increased azimuth/lateral resolution.
2. The peak sidelobe level expresses the array's ability to suppress energy coming from directions off of the main response axis, and is highly dependent on the number of and placement of elements, in addition to the excitation amplitude and phase (also referred to as the (complex) weight of each element). A low peak sidelobe level is advantageous for imaging a point target in a non-reflecting background.

²Grating lobes are copies of the mainlobe, which appear in the beampattern of periodic arrays when the interelement distance is greater than the minimum given from the spatial sampling theorem.

3. The energy in the sidelobe region of the beampattern is also of great interest, since it relates to the contrast resolution of the array. The contrast resolution expresses the array's ability to separate two scatterers of differing strength in close proximity.

Various applications have different requirements, and a substantial part of the job of an array designer is thus to find the amplitude weights, phase weights and placement of the array elements which are optimal with respect to the application of interest. There are other parameters that are also vital to imaging performance. One in particular is the bandwidth of the emitted wavefield, which is proportional to the range/axial resolution of the imaging system. The frequency range of the emitted pulse is also an important design choice since the penetration depth of a wavefield is inversely proportional to its frequency. The transmitted pulse type (e.g. narrowband, frequency modulated) also plays part here, where the choice of pulse type results in an ambiguity between the range/axial resolution and the Doppler capabilities of the system.

Two fundamental problems in array design are finding the:

1. Optimal (complex) weights³ to apply to the array elements.
2. Optimal placement of the array elements.

What is meant by optimal varies greatly depending on the application, and the requirements and constraints it puts on the design process, though by large it is related to the resolution measures mentioned above. The first of these problems is referred to as *array pattern synthesis*, and is discussed in a section further down.

To better understand how beamforming fits into the complete imaging process, we now look at a model for a typical processing chain for image generation. A block diagram of the processing chain is shown in Figure 4. Note that this processing chain works equally well on narrowband transmission as on wideband transmission. In the first step the received signals are digitized through an analog-to-digital (A/D) converter. The most important requirement at this stage is that the Nyquist-Shannon sampling theorem must be satisfied, i.e. that the sampling frequency must be at least twice the total bandwidth of the received signal. The Discrete Fourier Transform (DFT) is then taken on the data from each of the channels, by applying the Fast Fourier Transform (FFT). This takes the data over to the frequency domain. The data from each channel are then multiplied by weights, usually in order

³In the design process the optimization of complex weights encompasses both amplitude and phase control of the array elements.

to achieve sidelobe reduction. In the next step the data are beamformed over the desired angles in the resulting image. The frequency domain equivalent of Equation (1) for a frequency ω , assuming that we are looking in the far-field, is:

$$Y(\omega) = \sum_{m=1}^M Y_m(\omega) e^{-j\omega\Delta_m}.$$

Here $Y_m(\omega)$ is the DFT of the digitized signal from element m at frequency ω , and Δ_m is the time-delay needed at element m in order to achieve beamforming in a given direction. Note that for wideband signals this operation must be performed over the whole bandwidth. In the next step, the inverse DFT is then taken on the beamformed data, transforming the data back to the time domain. Range intensity for each of the beams can then be extracted through e.g. matched filtering with the excited pulse, which is simply a correlation of the received signal by that of the excited pulse. The image can then be displayed with intensities which are proportional to the output from the matched filter.

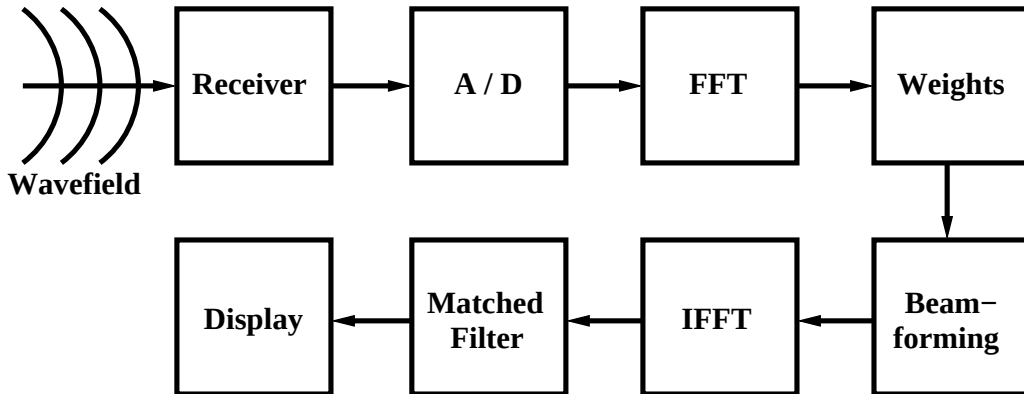


Figure 4: A typical imaging processing chain, containing a receiver, an analog-to-digital converter (A/D), the multiplication of weights, the Fast Fourier Transform (FFT), beamforming, the inverse FFT (IFFT), a time-domain matched filter, and finally the display.

Even though the imaging processing chain outlined above contains many steps, the underlying beamforming theory still gives us a very good picture of the qualities in the resulting image. The analysis of beamforming systems is usually performed by considering the scattered field from an ideal point reflector. Assuming linearity a reflecting structure can then be modelled by a collection of such point reflectors. To investigate this further we consider

an example from sonar. Though, the ideas transfer directly to medical ultrasound imaging as well. A system having a single element transducer and a 128 element uniform linear hydrophone array is considered. The excitation pulse is a linear frequency-modulated waveform, with a center frequency of $f_0 = 100$ kHz and a bandwidth of 10 kHz. The speed of sound is $c = 1500$ m/s, and the duration of the pulse is $t_p = 10$ ms. A point target has been placed 100 m directly in front of the uniform linear hydrophone array. Figure 5 shows the resulting intensity plot, in polar coordinates. Even though the intensity of the target (mainlobe) is well above the intensity of the sidelobes, the sidelobes are still clearly visible as a half circle continuation of the mainlobe. It is important to note here that it is the finite support of the transmitted pulse which has allowed us to extract position in both range and angle. If a continuous-wave pulse had been transmitted the target could only have been positioned in angle. Continuous-wave pulsed excitation is used for velocity estimation of targets, by extracting the Doppler shift.

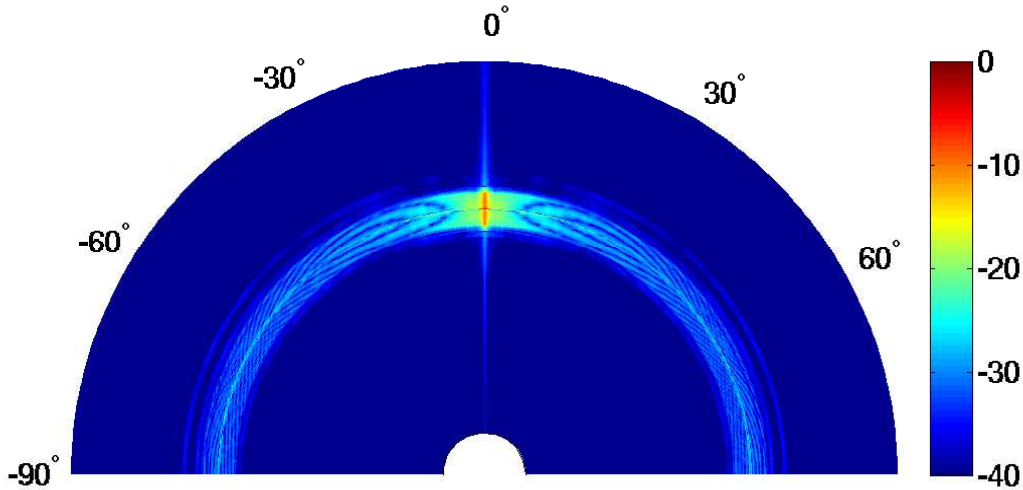


Figure 5: Intensity plot shown in polar coordinates. There is a target located at 0° , with a range of $2/3$ of the total plotted range. Uniform weighting has been applied to the data during beamforming. The intensity is shown in decibel.

Further details of the image in Figure 5, around the range of the target, can be seen in Figure 6. On the left we see that it is the temporal extent of the pulse which determines the range length of the mainlobe ($c \cdot t_p / 2 = (1500 \text{ m/s} \cdot 10 \text{ ms}) / 2 = 7,5 \text{ m}$) on each side of the actual target. On the right we see a plot showing the intensities at exactly the target range. In a

section below we will see clearly how shaping of the beampattern directly affects both the width of the mainlobe and the sidelobe levels.

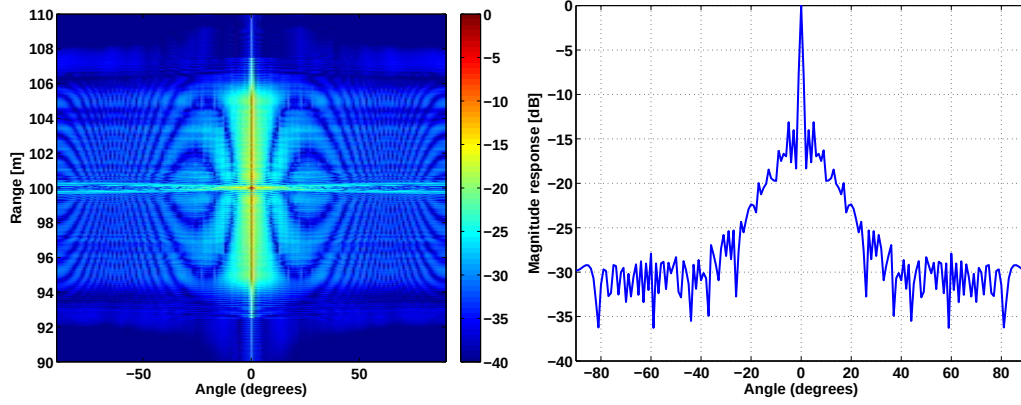


Figure 6: Intensity plot in cartesian coordinates around a target located at 0° , with a range of 100m from the receiver (left), shown in decibel. Also shown is a plot of the intensity at 100m from the receiver as a function of angle (right).

We have now touched the surface of echo-imaging systems, and seen how beamforming is utilized in such systems. For an in-depth look at beamforming the array signal processing textbooks [2, 3] are worth looking into, while brief accounts are offered in the survey articles [4, 5]. In the following two sections we will now take a closer look at two principal applications of beamforming for imaging, namely sonar and medical ultrasound.

Sonar

Sonar is an acronym for **s**ound **n**avigation **a**nd **r**anging. Sonar systems are used to detect, locate and/or classify objects in the sea by the use of acoustic waves. Sonars operate in one of two modes. In active mode an acoustic signal is transmitted, which is referred to as pinging, and the received echo is then analyzed. Though, unlike in medical ultrasound, sonar systems may also operate in passive mode, where it is the target's self noise which is analyzed. This self noise can be anything from a vessel's engine to the almost lyrical sounds made by whales. The performance of active sonars is by large limited by the energy reflected to the receiver, while the performance of passive sonars is largely limited by the lack of knowledge of the emitted sounds being analyzed.

The history of sonar started in 1912, less than a month after the tragic sinking of the Titanic. English meteorologist Lewis Richardson took out a patent for an iceberg detection system based on echo ranging. However, at this time the technology for realizing such a system didn't exist. It was only during World War I that hydrophones were developed, at that time for detecting submarines. During World War II the emergence of radar technology, with its 360° imaging, had synergy effects with the sonar community and sonar technology was taken beyond that of simply echo ranging to actual imaging. It is worth noting that sonars differ greatly from radars in one important aspect, since they are affected by the relatively large variations in the propagation characteristics of the underwater medium. During the cold war sonar became an essential tool for detecting (and classifying) submarines, both in active and passive mode. During the same period, sonars were also developed to become an important tool in fishery applications and marine biology. Since then development has surpassed the expectations of even the most foresighted, and sonar is currently used across almost all maritime disciplines, in some form or another. A nice introduction to the technicalities of sonar can be found in [6].

One typical setup for sonar is shown in Figure 7. The sonar is mounted on the hull of the vessel and insonifies a (360°) volume around it. In noise sensitive applications the hydrophones can be mounted on a towed antenna instead, to avoid the ship's self noise from e.g. the propeller or its engine. Other typical setups include mounting the sonar on a fixed underwater location or on buoys. The operating frequency of sonar systems can vary greatly, from just a few kilohertz for applications requiring long range or sub bottom penetration, to several hundred kilohertz for applications requiring high range resolution. To ensure penetration and good range estimates, frequency modulated waveforms are becoming more and more common. One preferred waveform is the hyperbolic frequency modulated one, given its name because it sweeps linearly between the reciprocal of the lower and upper frequencies. A problem with linear frequency modulated pulses is that there is a great degree of signal loss if the target is moving, due to the Doppler effect. The hyperbolic frequency modulated pulse owes much of its popularity to its Doppler-invariant property. Sonar continues to be an active field of research, facing many challenging problems. High resolution and adaptive beamforming [7] are currently very active areas of research, and challenging media such as shallow water are receiving their share of attention. Another application under intense investigation is acoustic communication [8]. Like imaging, it is greatly affected by the varying propagation characteristics of the water medium.

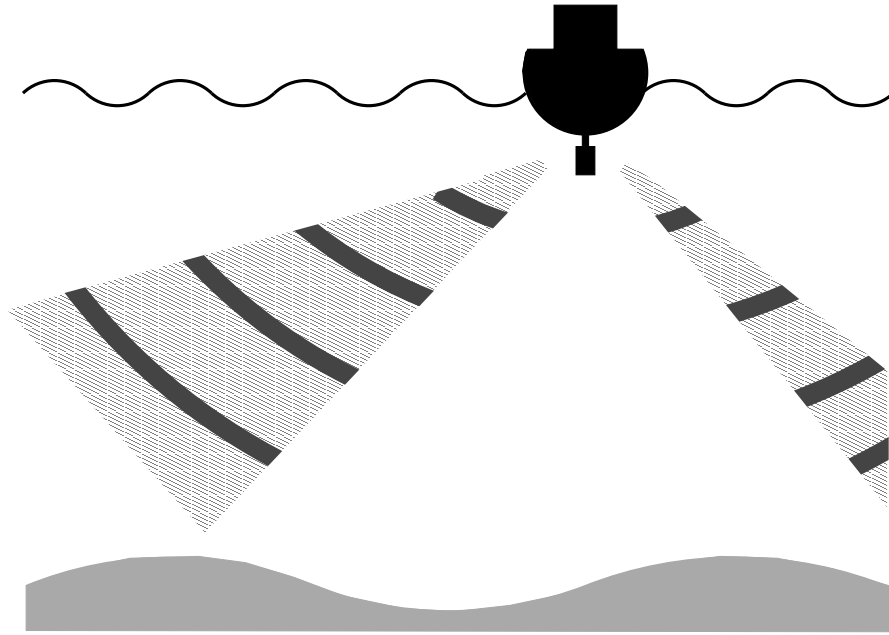


Figure 7: One typical setup for sonar imaging, with the array mounted below the hull of the vessel (seen from behind), imaging the surrounding volume.

Medical Ultrasound

The history of medical ultrasound goes back more than 50 years, when tests were started using modified sonar equipment. It was seen that the principles of sonar and radar could be used to image human tissue, whose consistence is not that different from water (this is not so suprising since live tissue has a high water content). The first ultrasound systems having diagnostic value displayed what came to be known as A-mode images, where the “A” stands for *amplitude*. The A-mode technology had no focusing, and simply displayed a one-dimensional signal giving the echo strength. In the 1950s and 1960s the B-mode technology was developed, with the “B” standing for *brightness*, giving the first two-dimensional views of the body. The B-mode technology forms the basis of the technology which today permeates most modern medical facilities. In a B-mode display the brightness in the image is proportional to the echo strength. In the beginning the B-mode images were generated using mechanically moving transducers, so that scans in various directions could be synthesized into an image. However, in the mid 1960s the first electronically steered array transducers were introduced, and this is the technology which has transformed into todays advanced real-time scanners.

Figure 8 shows how B-mode pictures are acquired, while Figure 9 shows a snapshot of a fetus taken on a B-mode scanner. An important difference between medical ultrasound imaging and most sonar systems is that beam-forming is done at both transmission and reception in medical ultrasound imaging, while sonar systems insonify a large sector in order to achieve a satisfactory frame rate. Today’s “hot” technology is by far 3D imaging using 2D arrays, which allows views of the inner body from any plane. The first commercial 3D ultrasound imaging systems have already been in production for a few years. Though, 3D imaging still faces several challenges which must be overcome, such as achieving a greater framerate as well as a higher signal-to-noise ratio (through a greater channel count). Finally, it is worth mentioning one of the great leaps taken in medical ultrasound imaging, namely harmonic imaging. It was seen that a clearer image could be synthesized by processing the second harmonic frequency instead of the frequency of the emitted pulse. See [9] for a discussion of the developments of this technology, as well some ideas on future developments. Reference [10] gives a nice overview of medical ultrasound imaging in general. Finally, one major difference worth noting between medical ultrasound imaging systems and sonars, are the transducers. Modern medical ultrasound systems have a very high relative bandwidth, so that the emitted pulse can be just a few wavelengths long, giving high range resolution. In addition, the development of promising technologies such as capacitive micromachined ultrasonic transducers (cMUT) [11] as an alternative to the currently used piezoelectric materials will most likely bring great changes to medical ultrasound imaging systems.

Having considered two main applications of interest with respect to beam-forming for imaging, we now turn to one of the fundamental problems in array design.

Array Pattern Synthesis

Arrays of elements are used in a wide range of applications where we try to “pick up” information from the surrounding space. As mentioned previously, there are two fundamental problems when designing an array, one of which is to find the optimal (complex) weights to apply to the array elements. This is also referred to as *array pattern synthesis*, or apodization. The goal is to find the magnitude weights and time-delay which should be applied to each of the elements so that the beampattern equals or approximates some desired beampattern. To better understand this problem in the context of imaging, compare the left-hand side of Figure 6 with the left-hand side of Figure 11.

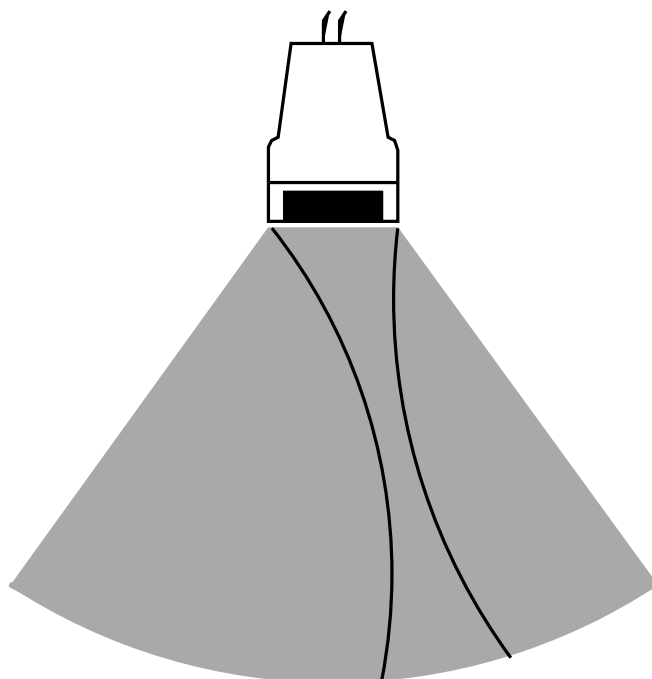


Figure 8: Ultrasound transducer acquiring B-mode images.

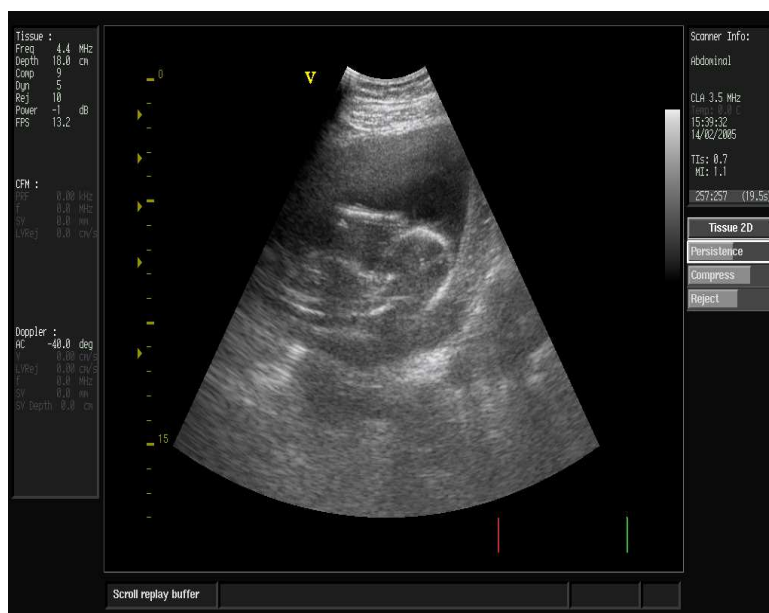


Figure 9: B-mode ultrasound image of fetus, recorded on a GE VingMed Ultrasound System FiVe scanner.

In the first figure we see the image with uniform weights applied to the data during beamforming. In the second figure we see the same data, but with a Hamming-window applied to the data during beamforming. Visually it is easy to see that the band of sidelobes has been greatly suppressed in the latter compared to the first figure, though at the expense of the target appearing wider in the bearing direction in the latter figure. This illustrates the trade-off when choosing a window, between precise localization of the target and of noise reduction from other directions (as mentioned above, in connection with the time-frequency uncertainty principle).

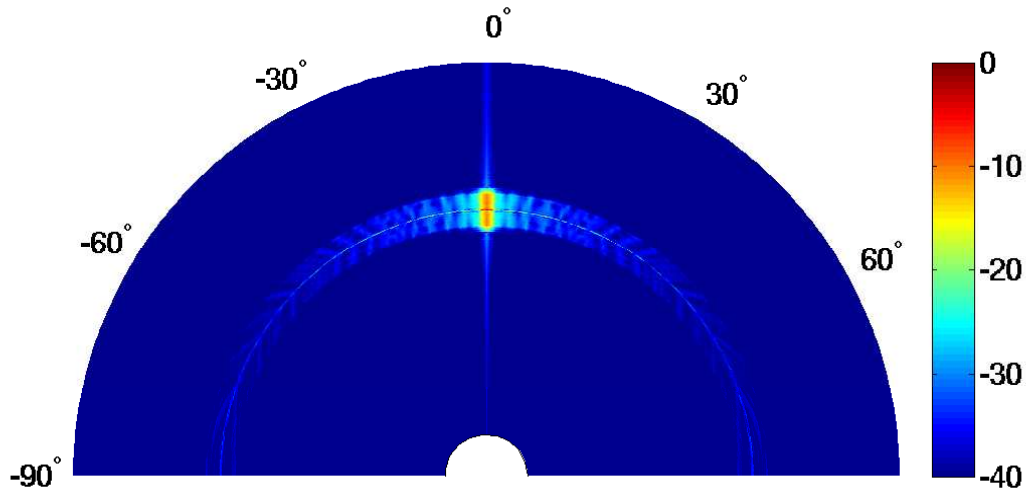


Figure 10: Intensity plot shown in polar coordinates. There is a target located at 0° , with a range of $2/3$ of the total plotted range. A Hamming-window has been applied to the data during beamforming. The intensity is shown in decibel.

There are several measures of performance relating to image quality, with the most common being beamwidth and sidelobe levels. Sidelobe levels are usually optimized with respect to the $\mathcal{L}_2$ norm or the $\mathcal{L}_\infty$ norm. The former gives the sidelobe energy, while the latter is the peak sidelobe level. From a theoretical viewpoint, optimization of the peak sidelobe level is mathematically more challenging. However, with respect to discernibility of artifacts in the synthesized images a low sidelobe energy is of essence. Both optimization of the peak sidelobe level as well as sidelobe energy are treated in this thesis.

The process of beamforming can be thought of as temporal filtering with

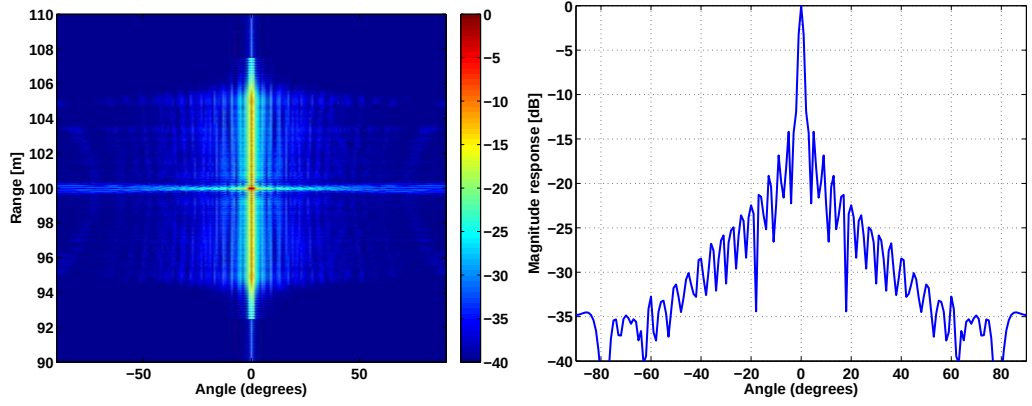


Figure 11: Intensity plot in cartesian coordinates around a target located at 0° , with a range of 100 m from the receiver (left), shown in decibel. Also shown is a plot of the intensity at 100 m from the receiver as a function of angle (right).

the added dimension of space, and is often referred to as *space-time processing*. It is therefore not surprising that many of the ideas in array pattern synthesis stem from FIR filter window design. Fundamental in finding the optimal filter in the Chebyshev sense was the work done by Remez. In [12], the *Remez exchange algorithm* was shown to find the best polynomial approximation to a given function, assuming that it satisfies some conditions. In [13], Parks and McClellan managed to show that the optimal coefficients in the Chebyshev sense can be found using the Remez exchange algorithm. This algorithm for filters thus became known as the *Parks-McClellan algorithm*. The ideas for finding the Chebyshev weights for temporal filters transfer to arrays, which was shown in the now famous paper from 1946 by Dolph [14]. In it the solution was found for the weights that minimize the peak sidelobe level (i.e. minimize the $\mathcal{L}_\infty$ norm) over the sidelobe region) for a linear array given a specific beamwidth.

Bibliography

- [1] D. Donnelly, “The Fast Fourier Transform for experimentalists, part IV: Chirp of a bat,” *Computing in Science & Engineering*, vol. 8, pp. 72–78, Mar.-Apr. 2006.
- [2] D. H. Johnson and D. E. Dudgeon, *Array Signal Processing: Concepts and Techniques*. Englewood Cliffs, NJ: Prentice-Hall, 1st ed., 1993.
- [3] H. L. Van Trees, *Optimum Array Processing*. Detection, Estimation, and Modulation Theory, New York, NY: Wiley, 1st ed., 2002.
- [4] B. D. Van Veen and K. M. Buckley, “Beamforming: A versatile approach to spatial filtering,” *IEEE Signal Processing Mag.*, vol. 5, pp. 4–24, Apr. 1988.
- [5] H. Krim and M. Viberg, “Two decades of array signal processing research,” *IEEE Signal Processing Mag.*, vol. 13, pp. 67–94, July 1996.
- [6] R. O. Nielsen, *Sonar Signal Processing*. Norwood, MA: Artech House, 1st ed., 1991.
- [7] K. W. Lo, “Adaptive array processing for wide-band active sonars,” *IEEE J. Oceanic Eng.*, vol. 29, pp. 837–846, July 2004.
- [8] D. B. Kilfoyle and A. B. Baggeroer, “The state of the art in underwater acoustic telemetry,” *IEEE J. Oceanic Eng.*, vol. 25, pp. 4–27, Jan. 2000.
- [9] P. A. Lewin, “Quo vadis medical ultrasound?,” *Ultrasonics*, vol. 42, pp. 1–7, Apr. 2004.
- [10] J. A. Jensen, “Medical ultrasound imaging,” *Prog. Biophys. Mol. Biol.*, vol. 93, pp. 153–165, Jan. 2007.
- [11] . Oralkan, A. S. Ergun, J. A. Johnson, M. Karaman, U. Demirci, K. Kaviani, T. H. Lee, and B. T. Khuri-Yakub, “Capacitive micromachined ultrasonic transducers: Next-generation arrays for acoustic imaging,”

IEEE Trans. Ultrason., Ferroelect., Freq. Contr., vol. 49, pp. 1596–1610, Nov. 2002.

- [12] E. Y. Remez, “General computational methods of Chebyshev approximation,” *Atomic Energy Comission Translation 4491*, pp. 1–85, 1957.
- [13] T. W. Parks and J. H. McClellan, “Chebyshev approximation for non-recursive digital filters with linear phase,” *IEEE Trans. Circuit Theory*, vol. 19, pp. 189–194, Mar. 1972.
- [14] C. L. Dolph, “A current distribution for broadside arrays which optimizes the relationship between beam width and side-lobe level,” *Proc. IRE*, vol. 34, pp. 335–348, June 1946.